
ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Department of Information Technology Synopsis On NyayaMitra : AI Legal Assistant

1.INTRODUCTION:

1.1 Overview

NyayaMitra is an advanced Artificial Intelligence based web application designed to provide educational legal assistance based on Indian constitutional and legal knowledge. In today's digital era, access to legal information is often limited due to complex language, lack of awareness, and scattered sources of legal data. Common users face difficulty in understanding legal rights, procedures, and laws because most legal documents are written in technical language that is difficult for non-professionals to interpret.

One of the key features of NyayaMitra is that it can operate even without external API keys, functioning in a context-only mode where it retrieves and displays relevant legal information without generating additional responses. This makes the system highly reliable for demonstration and educational purposes.

1.2 Purpose

The main purpose of the NyayaMitra system is to provide a simplified and structured platform for users to access legal knowledge in an easy and understandable manner. The project aims to bridge the gap between complex legal systems and general users by converting technical legal information into natural language explanations.

This system is particularly useful for students, beginners, and individuals who do not have formal legal training but want to understand their rights and responsibilities. Overall, the purpose of NyayaMitra is to make legal information more accessible, promote awareness, and demonstrate the practical implementation of AI-based systems.

2. LITERATURE SURVEY

2.1 Existing Problem

In the current scenario, accessing legal information is a challenging task for most individuals:

- Legal information related to the Indian Constitution is complex and difficult for common people to understand.
- People do not have easy access to reliable and structured legal guidance for everyday situations.
- Most online sources provide scattered or incomplete information, making it hard to find accurate answers quickly.
- Consulting a lawyer for basic queries is often time-consuming and expensive , especially for simple legal doubts.

2.2 Proposed Solution

To overcome the limitations of existing systems, NyayaMitra proposes

- **RAG-Based Legal Information Retrieval** : NyayaMitra uses Retrieval-Augmented Generation to fetch relevant legal content from a vector database. This ensures that responses are grounded in real legal data instead of generating answers blindly.
- **Context-Aware Chat System** : The system provides a chat interface where users can ask legal questions in natural language. It retrieves context from stored data and generates meaningful responses, improving understanding of complex legal topics.
- **Structured Action Plan Generation** : NyayaMitra includes a feature that converts user queries into a structured legal guidance plan.This helps users understand steps, precautions, and legal framework in a simplified and organized manner.
- **Offline-First and Flexible Architecture** : The system works even without external API keys by providing context-only responses. This makes it reliable for demos, educational use, and environments with limited internet or API access.

3.THEORATICAL ANALYSIS

3.1 Block Diagram

The overall working of NyayaMitra can be explained using a block diagram consisting of the following components:

User Interface (Frontend), API Layer (Backend), Retrieval Module, Vector Database (ChromaDB)

Language Model (Optional), Response Generator

NyayaMitra – System Block Diagram

AI Legal Assistant using RAG

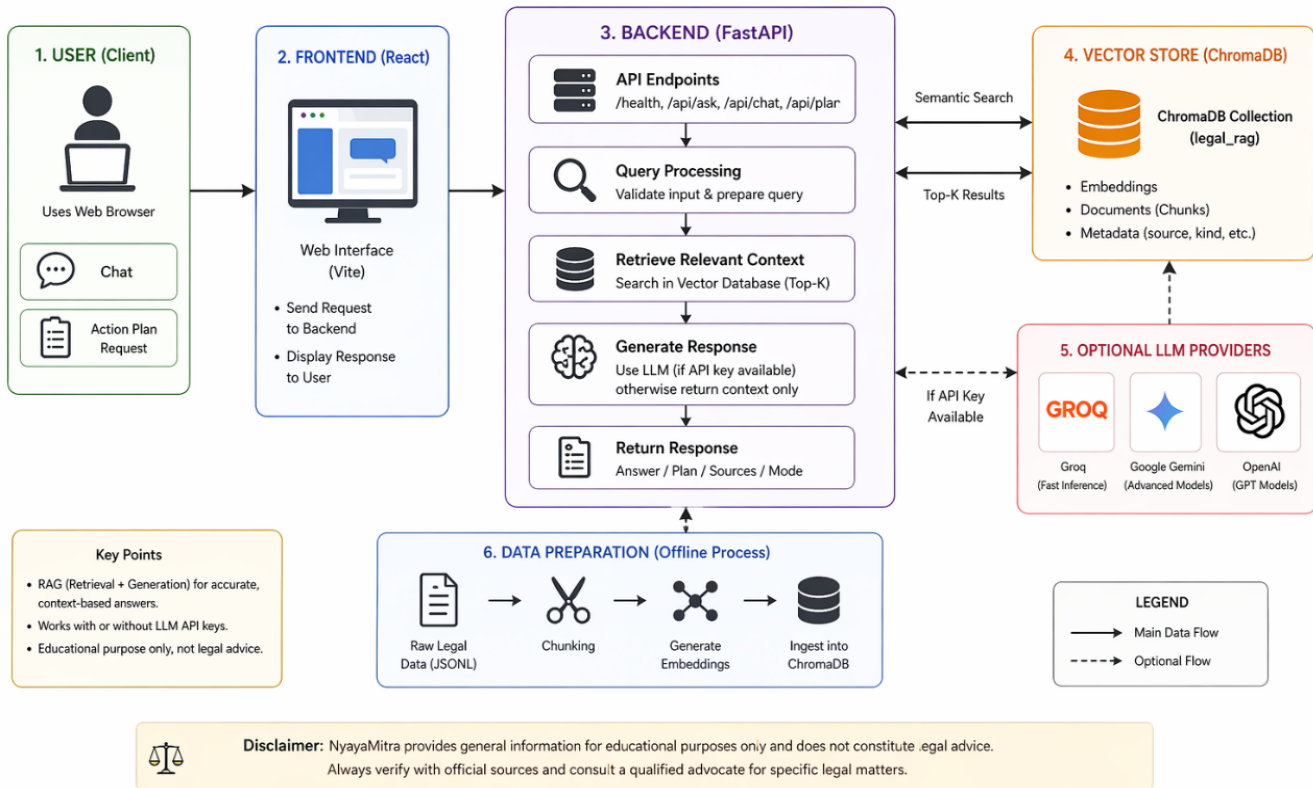


Fig.1. Block Diagram of NyayaMitra

3.2 Hardware/Software Designing

3.2.1 Hardware Requirements:

The system can run on a standard computer with minimum specifications such as:

- **Processor:** Intel i5 or equivalent
- **RAM:** 8 GB or higher
- **Storage:** 256 GB
- **Internet** connection (for initial setup and API usage)
- **Monitor :** Full HD(1920x1080) 14-inch or larger
- **Server :** Quad-core processor with 8GB RAM for hosting .

3.2.2 Software Requirements:

- **Programming Languages:**
 - Python (for backend development)
 - JavaScript (for frontend development)

- **Backend Requirements:**
 - Python (Version 3.11 or above)
 - FastAPI (API development)
 - Uvicorn (ASGI server to run backend)
- **Frontend Requirements:**
 - Node.js (Version 18+)
 - React (Frontend framework)
- **Database / Storage:**
 - ChromaDB (for storing embeddings and chunks)
- **AI / Machine Learning Libraries:**
 - LangChain (for RAG pipeline)
 - Sentence Transformers (for embeddings generation)
- **Development Tools:**
 - Visual Studio Code (recommended IDE)
 - Git (for version control)
 - Postman (for API testing)

4.APPLICATIONS

- **Legal Education Support :** NyayaMitra helps students understand constitutional and legal concepts in a simplified manner. It provides context-based explanations, making complex legal topics easier to learn.
- **General Public Awareness :** The system can be used by common people to get basic knowledge about their legal rights. It improves awareness about laws without needing deep legal expertise.
- **Legal Research Assistance:** NyayaMitra assists in quickly retrieving relevant legal information from large datasets. This helps users save time in searching and understanding legal documents.
- **Structured Legal Guidance :** It generates step-by-step action plans for legal situations. This helps users understand what actions they can take in a clear and organized way.
- **Demo and Educational AI System :** The project can be used as a demonstration of modern AI systems using Retrieval-Augmented Generation. It is useful for learning how real-world AI applications are built.

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